DebateSpeak AI — Multi-Sprint Engineering Task Breakdown & Role Matrix (TASKS.md)

Field	Value
Project Name	**DebateSpeak AI**
Target Timeline	10–12 Weeks across **5 Sprints** (Mapped directly to **PA1–PA5**)
Team Size	5 Engineers
Companion Documents	SOFTWARE_ARCHITECTURE_SPEC.md, TESTING_PLAN.md

1. Team Role Allocation Matrix

Performed by: Nguyễn Đình Thiên Lộc (24125093) | Reviewed by: Nguyễn Bảo Minh Triết (24125047) | Edited by: Trần Lê Anh Tuấn (24125107)

Role ID	Role Title	Member	Candidate Technologies (Tentative / Subject to Change)
R1	**Frontend & WebRTC/Audio Lead**	Nguyễn Bảo Minh Triết (`24125047`)	React, TypeScript, Tailwind CSS, LiveKit Client SDK
R2	**Backend & Real-Time Media Lead**	Nguyễn Hồng Tấn Tài (`24125078`)	FastAPI, SQLAlchemy, Alembic, PostgreSQL, Redis, LiveKit API
R3	**AI Audio & Speech Pipeline Lead**	Nguyễn Anh Khoa (`24125058`)	Python, LiveKit Agents, Silero VAD, STT Adapter Protocol
R4	**AI Analysis & Agent Orchestrator Lead**	Trần Lê Anh Tuấn (`24125107`)	Python, Pydantic v2, LLM Adapter Protocol, Redis Queue (ARQ)
R5	**DevOps & QA / Fact-Checking Lead & Release Custodian**	Nguyễn Đình Thiên Lộc (`24125093`)	Docker Compose, Caddy (TLS), GitHub Actions, Search Adapter, Playwright, pytest

> **Note on Implementation Status & Technology Choices:** Because active implementation has not yet started in Sprint 1 (conception & proposal phase), primary ownership of code modules is not assigned in advance. The candidate technologies listed above reflect initial architectural planning and remain tentative and subject to change based on prototyping, research spikes, and Spec Kit specifications in future sprints. Furthermore, for Sprint 1 documentation, Nguyễn Đình Thiên Lộc served as Documentation Custodian & Git Release Manager to manage consolidated repository commits on behalf of all collaborating authors.

2. Multi-Sprint Work Breakdown Structure (5 Sprints = PA1 to PA5)

Performed by: Nguyễn Đình Thiên Lộc (24125093) | Reviewed by: Nguyễn Hồng Tấn Tài (24125078) | Edited by: Nguyễn Anh Khoa (24125058)

Sprint 1 (Weeks 1–2 / PA1): Foundation, Proposals & Setup

Sprint Goal: Comprehensive project proposal (10 FGs), existing app survey, team contract, dev environment setup, and baseline monorepo with CI.

Task ID	Task Description	Assignee	Reviewer	Deliverable / Output	Status
T1.1	Repository setup, monorepo layout, code style linters (Ruff, ESLint), GitHub Actions CI	**R5**	**R2**	`.github/workflows/ci.yml`	Done
T1.2	Docker Compose setup for local dev (PostgreSQL 16, Redis 7, LiveKit dev) & clean `.env.example`	**R5**	**R2**	`docker-compose.yml`, `.env.example`	Done
T1.3	Author Project Proposal (10 Functional Groups & 2 Actors)	**R1**	**R5**	`docs/requirements/PROJECT_PROPOSAL.md`	Done
T1.4	Conduct Existing App Survey (ArguFight, Khaos Live, ELSA Speak)	**R2**	**R4**	`docs/requirements/EXISTING_APP_SURVEY.md`	Done
T1.5	Draft Team Contract & Code Standards	**R3**	**R1**	`docs/management/TEAM_CONTRACT.md`	Done
T1.6	Setup Dev Tools, Scrum Process & Incremental Slices	**R5**	**R2**	`docs/management/DEV_TOOLS_AND_PROCESS.md`	Done
T1.7	Design System Architecture Spec & PostgreSQL Schema	**R4**	**R5**	`docs/architecture/SOFTWARE_ARCHITECTURE_SPEC.md`	Done
T1.8	Draft QA & Testing Plan	**R5**	**R3**	`docs/testing/TESTING_PLAN.md`	Done
T1.9	Configure `latexmk` academic report compilation & compile PDF	**R5**	**R1**	`report/report.tex`, `report/build/report.pdf`	Done

Sprint 2 (Weeks 3–4 / PA2): Project Planning, Vision Document & Build 1

Sprint Goal: Initial Project Plan, Vision Document, Spec Kit initialization in `/src`, and Build 1 (Auth + Room Lobby skeleton).

Task ID	Task Description	Assignee	Reviewer	Deliverable / Output	PA Mapping
T2.1	Author Project Plan: Goals, deliverables, risks, 5-sprint schedule, build plan	**R5**	**R1**	`docs/pa2/PROJECT_PLAN.md`	PA2 Part A
T2.2	Author Vision Document: Problem statement, positioning table, 10 feature paragraphs, workflows	**R1**	**R4**	`docs/pa2/VISION_DOCUMENT.md`	PA2 Part B
T2.3	Spec Kit Initialization in `/src`: `constitution.md`, initial spec artifacts, team training evidence	**R4**	**R5**	`src/constitution.md`, training logs	PA2 Part C
T2.4	Sprint 2 AI Usage Report & Weekly Report (Agile meetings, Jira progress)	**R3**	**R1**	`docs/pa2/WEEKLY_REPORT.md`	PA2 Part D
T2.5	Build 1 Core Backend: PostgreSQL schema migrations, Argon2id auth, JWT rotation	**R2**	**R5**	`api/app/auth/`	Build 1
T2.6	Build 1 Core Frontend: Vite + React shell, login/register UI, CEFR level selection	**R1**	**R2**	`web/src/pages/auth/`	Build 1
T2.7	Build 1 Room Lifecycle: `create_room`, `join_room` via invite code (Lobby state machine)	**R2, R1**	**R5**	`api/app/rooms/`, `web/src/pages/rooms/`	Build 1

Sprint 3 (Weeks 5–6 / PA3): Use-Case Modeling & First Functional Slice (Build 2)

Sprint Goal: Comprehensive use-case model with Mermaid diagrams, use-case specifications, UI prototypes, and full-stack implementation of 1 Functional Group.

Task ID	Task Description	Assignee	Reviewer	Deliverable / Output	PA Mapping
T3.1	Revised Project Plan & Vision Document with `Changes.md` (addressing TA feedback)	**R5**	**R1**	`docs/pa3/Changes.md`, revised docs	PA3 Part A, B
T3.2	Author Use-Case Model: Mermaid diagrams for all 10 FGs, actors, relationships	**R4**	**R2**	`docs/pa3/USE_CASE_MODEL.md`	PA3 Part C
T3.3	Author Use-Case Specifications: Detailed flows, pre/post conditions, and UI mockups for every UC	**R1, R4**	**R5**	`docs/pa3/USE_CASE_SPECIFICATIONS.md`	PA3 Part D
T3.4	**Implement 1 Functional Group (FG-03: Room & Live Session Management)** end-to-end using Spec Kit	**R2, R1**	**R5**	Full-stack working code	PA3 Part E (Build 2)
T3.5	Record narrated video demo of FG-03 and upload to YouTube	**R1**	**All**	YouTube Video Link	PA3 Part E
T3.6	Sprint 3 AI Usage Report & Weekly Report (Scrum minutes, Jira tracking)	**R3**	**R1**	`docs/pa3/WEEKLY_REPORT.md`	PA3 Part F

Sprint 4 (Weeks 7–8 / PA4): Software Architecture & Second Implementation Slice (Build 3)

Sprint Goal: Complete C4 architecture model (Context, Container, Component, Deployment diagrams), and full-stack implementation of 2 additional Functional Groups (FG-04 Audio + FG-05 STT Pipeline).

Task ID	Task Description	Assignee	Reviewer	Deliverable / Output	PA Mapping
T4.1	Revised Use-Case Specifications with `Changes.md`	**R4**	**R1**	`docs/pa4/Changes.md`, revised specs	PA4 Part A
T4.2	C4 Level 1: System Context Diagram & written explanation	**R4**	**R5**	`docs/pa4/ARCHITECTURE_C4.md`	PA4 Part B
T4.3	C4 Level 2 & 3: Container Diagram & Component Diagrams (API, Web, LiveKit, Workers)	**R4, R2**	**R5**	`docs/pa4/ARCHITECTURE_C4.md`	PA4 Part C
T4.4	Deployment Diagram: Docker Compose, Caddy TLS reverse proxy, cloud media mapping	**R5**	**R2**	`docs/pa4/DEPLOYMENT_DIAGRAM.md`	PA4 Part D
T4.5	**Implement 2 Functional Groups (FG-04 Real-Time Audio + FG-05 Streaming STT)** using Spec Kit	**R3, R1, R2**	**R5**	Full-stack working code	PA4 Part E (Build 3)
T4.6	Record narrated video demo of audio and live transcription pipeline	**R3**	**All**	YouTube Video Link	PA4 Part E
T4.7	Sprint 4 AI Usage Report & Weekly Report	**R5**	**R1**	`docs/pa4/WEEKLY_REPORT.md`	PA4 Part F

Sprint 5 (Weeks 9–10 / PA5): Comprehensive Testing, Hardening & Final Product Demo (Build 4)

Sprint Goal: Execute test plan, refine Spec Kit test cases, golden-set evaluations, final reflective report, and live 15-minute product presentation.

Task ID	Task Description	Assignee	Reviewer	Deliverable / Output	PA Mapping
T5.1	Test Plan & Refined Test Cases: Validate auto-generated test cases, add negative edge cases	**R5**	**R2**	`docs/pa5/TEST_EXECUTION_REPORT.md`	PA5 Part A
T5.2	Test Execution & Defect Reporting: Document pass/fail matrix, link bug reports	**R5, R1**	**R4**	Bug reports & test matrix	PA5 Part A
T5.3	Multi-Agent Analysis Hardening: Complete Language Coach, Argument Coach, Fact-Checker & Judge	**R4, R5**	**R2**	Working multi-agent report	Build 4 (Release)
T5.4	Personalized Learning Report UI & Learner History Progress Charts	**R1**	**R4**	Report viewer & analytics dashboard	Build 4 (Release)
T5.5	Conduct 15-minute Live Final Product Demo showcasing all features (no slides)	**All (Lead: R1)**	**Faculty**	Live Demo Session	PA5 Part B
T5.6	Author Reflective Report: SDLC critique, tooling evaluation, individual contributions	**All (Lead: R5)**	**All**	`docs/pa5/REFLECTIVE_REPORT.md`	PA5 Part C
T5.7	Final Codebase Freeze & Package Submission (`PA5-Group01.zip`)	**R5**	**All**	Final submission archive	PA5 Part D

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3. Individual Member Workload Matrix

Performed by: Nguyễn Đình Thiên Lộc (24125093) | Reviewed by: Trần Lê Anh Tuấn (24125107) | Edited by: Nguyễn Bảo Minh Triết (24125047)

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| ROLE 1: FRONTEND & WEBRTC/AUDIO LEAD (Triết - 24125047)                                       |
| Primary Focus: UI/UX, WebRTC Audio Integration, Lobby & Live Debate, Learning Reports, Demos  |
| Tasks: T1.3, T1.9, T2.2, T2.6, T2.7, T3.3, T3.4, T3.5, T4.5, T5.2, T5.4, T5.5                |
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| ROLE 2: BACKEND & REAL-TIME MEDIA LEAD (Tài - 24125078)                                       |
| Primary Focus: FastAPI API, PostgreSQL Models, JWT Auth, Room State Machine, WebSockets, PubSub |
| Tasks: T1.4, T2.5, T2.7, T3.2, T3.4, T4.3, T4.5, T5.1, T5.3, T5.5                             |
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| ROLE 3: AI AUDIO & SPEECH PIPELINE LEAD (Khoa - 24125058)                                     |
| Primary Focus: LiveKit Bot Worker, Audio Track Hooks, Silero VAD, Streaming STT Adapters      |
| Tasks: T1.5, T2.4, T3.6, T4.5, T4.6, T5.5                                                     |
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| ROLE 4: AI ANALYSIS & AGENT ORCHESTRATOR LEAD (Tuấn - 24125107)                               |
| Primary Focus: Agent Prompts, Spec Kit Models, Language/Argument Coach, Consensus Judge       |
| Tasks: T1.7, T2.3, T3.2, T3.3, T4.1, T4.2, T4.3, T5.3, T5.4, T5.5                            |
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| ROLE 5: DEVOPS & QA / FACT-CHECKING LEAD (Lộc - 24125093)                                     |
| Primary Focus: Staging Deployment, CI/CD, Search Retrieval, Security, E2E Testing, latexmk    |
| Tasks: T1.1, T1.2, T1.6, T1.8, T1.9, T2.1, T3.1, T4.4, T4.7, T5.1, T5.2, T5.6, T5.7          |
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